

ECEN 662: Estimation & Detection Theory

Lecture 11: Uniformly Minimum-Risk Equivariant Estimation

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- Let us consider a *location family* in which $X = (X_1, \dots, X_n)$ follows a joint density of the form

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where the function f is fixed and known, and θ is the unknown *location parameter*

- Note that by the *change-of-variable* formula, we can write

$$X_i = U_i + \theta, \quad \forall i \in [n]$$

such that

$$(U_1, \dots, U_n) \sim f(u_1, \dots, u_n),$$

which is *independent* of θ

Location invariance

- Note that for the aforementioned location family, we have

$$f_{\theta+c}(x_1 + c, \dots, x_n + c) = f_{\theta}(x_1, \dots, x_n)$$

i.e., the density is *invariant* under the same amount of shift in data and parameter. In this case, the family is said to be *location invariant*

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- Similarly, a loss function $L(\theta, d)$ is *location invariant* if $L(\theta + c, d + c) = L(\theta, d)$. Note that this implies that the loss function is of the form

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- A decision problem is *location invariant* if both the family of distributions and the loss function are *location invariant*
- When our decision problem displays this sort of *invariance*, it is reasonable to constrain our estimator to *respect* these symmetries as well

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- From the design perspective, the *equivariance* restriction is extremely appealing because it helps to collapse the risk function into a *single* number
- (Lemma) If δ is a *location-equivariant* estimator for a *location-invariant* decision problem, then its bias, variance, and risk do *not* depend on θ

- To see this, note that

$$\mathbb{E}_\theta[\delta(X_1, \dots, X_n) - \theta] = \mathbb{E}_\theta[\delta(X_1 - \theta, \dots, X_n - \theta)] = \mathbb{E}[\delta(U_1, \dots, U_n)]$$

where $U_i = X_i - \theta$ and

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- Next, we shall first develop a characterization of *all* location-equivariant estimators

A characterization of location-equivariant estimators

- First note that if δ_0 is a *location-equivariant* estimator, then any other estimator δ is location equivariant if and only if

$$\delta = \delta_0 - U$$

where U is *location invariant*, i.e.,

$$U(x_1 + c, \dots, x_n + c) = U(x_1, \dots, x_n), \quad \forall c \in \mathcal{R}$$

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- Next, a statistic U is *location invariant* if and only if it is a function of $(Y_1, \dots, Y_{i-1})$, where $Y_i = X_i - X_n$ for $i \in [n-1]$. Note that the difference above is taken between i and n simply for convenience

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- Combining the above two observations, we may conclude that *every* location-equivariant estimator has the form

$$\delta(X_1, \dots, X_n) = \delta_0(X_1, \dots, X_n) - v(Y_1, \dots, Y_{i-1})$$

where δ_0 is a *reference* location-equivariant estimator, v is an arbitrary function, and $Y_i = X_i - X_n$

Finding the UMREE

- Given the aforementioned characterization and the fact that the risk is *independent* of θ , to find an UMREE, it suffices to find a function $v(Y)$ where $Y = (Y_1, \dots, Y_{i-1})$ to minimize the risk

$$R_\delta(0) = \mathbb{E}_0[\rho(\delta_0(X) - v(Y))]$$

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- If ρ is convex and not monotone, then an UMREE exists. If ρ is also strictly convex, then the UMREE is unique
- For the *squared-error* loss $\rho(t) = t^2$, it is straightforward to see that

$$v^*(Y) = \mathbb{E}_0[\delta_0(X)|Y]$$

so the unique MREE is given by

$$\delta^*(X) = \delta_0(X) - \mathbb{E}_0[\delta_0(X)|Y]$$

The Pitman estimator of location

- The following more explicit form of δ^* is known as the *Pitman estimator of location*:

$$\delta^*(X) = \frac{\int_{-\infty}^{\infty} u f(X_1 - u, \dots, X_n - u) du}{\int_{-\infty}^{\infty} f(X_1 - u, \dots, X_n - u) du}$$

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- To this end, consider $Z = (Y_1, \dots, Y_{n-1}, X_n)$ and denote its density by g_θ . We have

$$\mathbb{E}_0[X_n|Y] = \frac{\int_{-\infty}^{\infty} t g_0(Y_1, \dots, Y_{n-1}, t) dt}{\int_{-\infty}^{\infty} g_0(Y_1, \dots, Y_{n-1}, t) dt}$$

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- By the *change-of-variable* formula, we have

$$g_\theta(y_1, \dots, y_{n-1}, x_n) = f_\theta(y_1 + x_n, \dots, y_{n-1} + x_n, x_n)$$

- We thus have

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- Replacing t by $X_n - u$ gives

$$\begin{aligned} \mathbb{E}_0[X_n|Y] &= \frac{\int_{-\infty}^{\infty} (X_n - u) f(Y_1 + X_n - u, \dots, Y_{n-1} + X_n - u, X_n - u) du}{\int_{-\infty}^{\infty} f(Y_1 + X_n - u, \dots, Y_{n-1} + X_n - u, X_n - u) du} \\ &= \frac{\int_{-\infty}^{\infty} (X_n - u) f(X_1 - u, \dots, X_{n-1} - u, X_n - u) du}{\int_{-\infty}^{\infty} f(X_1 - u, \dots, X_{n-1} - u, X_n - u) du} \\ &= X_n - \frac{\int_{-\infty}^{\infty} u f(X_1 - u, \dots, X_{n-1} - u, X_n - u) du}{\int_{-\infty}^{\infty} f(X_1 - u, \dots, X_{n-1} - u, X_n - u) du} \end{aligned}$$

- We thus have

$$\mathbb{E}_0[X_n|Y] = \frac{\int_{-\infty}^{\infty} tf(Y_1 + t, \dots, Y_{n-1} + t, t)dt}{\int_{-\infty}^{\infty} f(Y_1 + t, \dots, Y_{n-1} + t, t)dt}$$

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- The UMREE is thus given by

$$\delta^*(X) = X_n - \mathbb{E}_0[X_n|Y] = \frac{\int_{-\infty}^{\infty} uf(X_1 - u, \dots, X_n - u)du}{\int_{-\infty}^{\infty} f(X_1 - u, \dots, X_n - u)du}$$

Example: Uniform with unknown center

- Suppose that $X_i \stackrel{iid}{\sim} \mathcal{U}\left(\left[\theta - \frac{b}{2}, \theta + \frac{b}{2}\right]\right)$, where b is known and θ is the unknown parameter

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- Suppose that $X_i \stackrel{iid}{\sim} \mathcal{U}([\theta - \frac{b}{2}, \theta + \frac{b}{2}])$, where b is known and θ is the unknown parameter
- The likelihood

$$f_{\theta}(x_1, \dots, x_n) = \frac{1}{b^n} 1_{\{\theta - \frac{b}{2} \leq X_{(1)} \leq X_{(n)} \leq \theta + \frac{b}{2}\}}$$

where

$$X_{(1)} = \min_{i \in [n]} X_i \quad \text{and} \quad X_{(n)} = \max_{i \in [n]} X_i$$

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- Thus, any

$$\delta(X) \in \left[X_{(n)} - \frac{b}{2}, X_{(1)} + \frac{b}{2} \right]$$

is an *MLE* of θ

- Under the squared-error loss, the *UMREE* is unique and is given by

$$\begin{aligned}
 \delta^*(X) &= \frac{\int_{-\infty}^{\infty} u \frac{1}{b^n} 1_{\{X_{(n)} - b/2 \leq u \leq X_{(1)} + b/2\}} du}{\int_{-\infty}^{\infty} \frac{1}{b^n} 1_{\{X_{(n)} - b/2 \leq u \leq X_{(1)} + b/2\}} du} \\
 &= \frac{\frac{1}{2}(X_{(1)} + b/2)^2 - \frac{1}{2}(X_{(n)} - b/2)^2}{(X_{(1)} + b/2) - (X_{(n)} - b/2)} \\
 &= \frac{X_{(1)} + X_{(n)}}{2}
 \end{aligned}$$

The unbiasedness of UMREE

- Note that in the previous example, the UMREE

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is *unbiased* – you should be able to convince yourself and mathematically prove it as well

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- (Lemma) If $\delta(X)$ is location equivariant with constant bias b , then $\delta'(X) = \delta(X) - b$ is unbiased, location equivariant, and has a *smaller* risk than $\delta(X)$

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- (Lemma) If $\delta(X)$ is location equivariant with constant bias b , then $\delta'(X) = \delta(X) - b$ is unbiased, location equivariant, and has a *smaller* risk than $\delta(X)$
- To see that $\delta'(X)$ has a *smaller* risk than $\delta(X)$, note that

$$\begin{aligned} R_{\delta'}(\theta) &= \mathbb{E}_0[(\delta'(X))^2] \\ &= \mathbb{E}_0[(\delta(X) - b)^2] \\ &= \mathbb{E}_0[(\delta(X))^2] + b^2 - 2b\mathbb{E}_0[\delta(X)] \\ &= R_\delta(\theta) + b^2 - 2b^2 = R_\delta(\theta) - b^2 \end{aligned}$$